1. Which of the following is the greatest value?

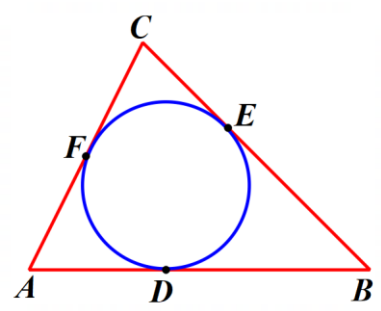
A) $\sqrt{\sqrt[3]{5\cdot 6}}$ **B)** $\sqrt{6\sqrt[3]{5}}$ **C)** $\sqrt{5\sqrt[3]{6}}$ **D)** $\sqrt[3]{6\sqrt{5}}$

2. Four numbers when added to three numbers combined at the same time results in 180, 197, 208 and 222. What are the four largest numbers?

A) 66 **B) 96** **C) 89** **D) 52**

3. A circle inscribed in ΔABC has side length $8\,cm$, $10\,cm$ and $12\,cm$, as shown in the figure.

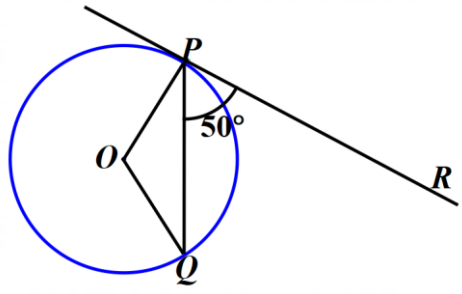
The length of the side BE is equal to:



A) 12 **B) 5** **C) 68** **D) 4**

4. In the figure, if **O** is the center of the circle, **PQ** is the chord, and the **PR** line touches the circle at **P** formed an angle of 50 ° with the chord PQ, then $\angle POQ$ is equal to:

A) 100° **B) 120°** **C) 112°** **D) 130°**



5. For a natural number, when it increases by 12, it is equal to 160 times of its multiplicative inverse, so that number is:

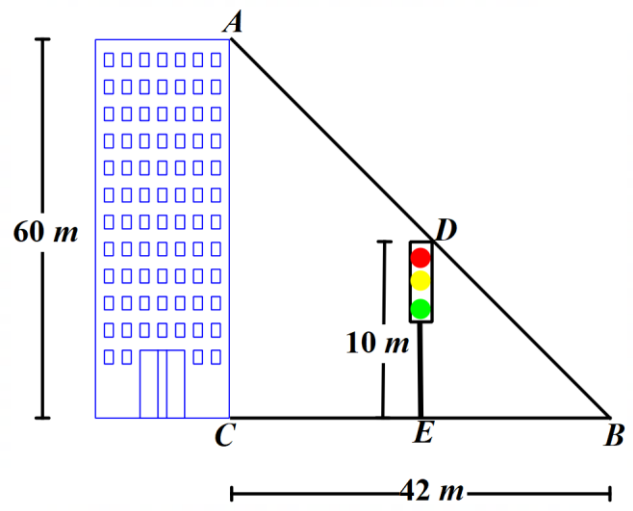
A) 26 **B) 8** **C) 6** **D) 32**

6. ***A*** takes 10 days less than ***B's*** time to complete a task. If ***A*** and ***B*** can work together to complete the task in 12 days, find time B alone to finish work.

A) 15 days **B) 18** days **C) 25** days **D) 30** days

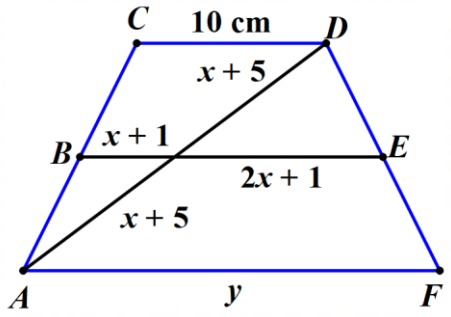
7. At a certain time of sunlight, a 60-meter-tall building casts a 42-meter shadow. How far is the shadow placed by the traffic light 10 meters high at the same time?

A) 6m **B) 12m** **C) 7m** **D) 5m**



8. Determine the length of the ***AD*** diagonal in the trapezoid below if ***CD*** || ***AF, B*** and ***E*** are the midpoints of ***AC*** and ***DF***, respectively.

A) 16cm **B) 8cm** **C) 27cm** **D) 18cm**

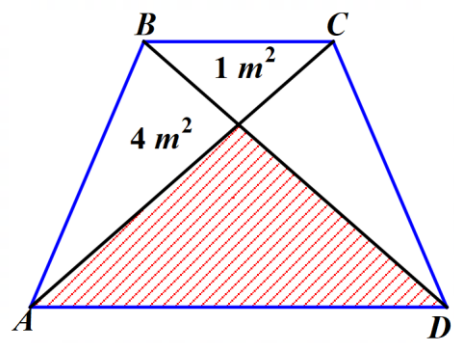


9. Let rectangle ***ABCD*** and ***O*** be the intersection of the diagonals ***AC*** and ***BD***. If the area of the triangle ***AOB***, ***BOC*** and ***COD*** is equal to $1\ m^2$, $0\ m^2$ and $2\ m^2$ respectively, thus, the value in m^2 of the area of the ***AOD*** triangle is:

A) $2\ m^2$ **B)** $6\ m^2$ **C)** $8\ m^2$ **D)** $3\ m^2$

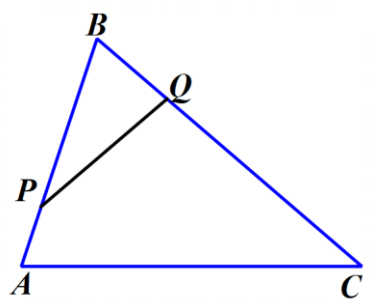
10. In the trapezoid **ABCD** below, find the area of the striped part.

A) 25 **B) 36** **C) 24** **D) 16**



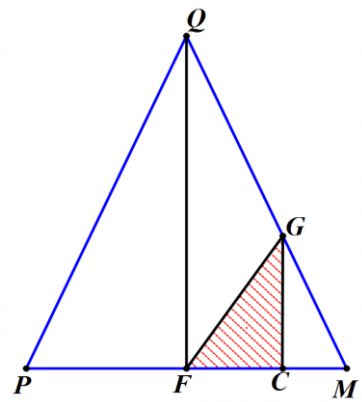
11. From the figure below, $PB = 4AP$, $\frac{BQ}{QC} = \frac{2}{7}$ and the area of the quadrilateral ***APQC*** is $37\ cm^2$. Find the area of the triangle ***BPQ***.

A) $10\ cm^2$ **B)** $8\ cm^2$ **C)** $4\ cm^2$ **D)** $12\ cm^2$



12. From the figure below, $\frac{FP}{FM}=\frac{1}{6}, [CG]//[QF], G$ is the center point of **MQ** and the area of triangle **PQM** is / . . **m²**. Find the area of striped part.

A) $\frac{150}{7}$ **B)** $\frac{300}{7}$ **C)** $\frac{115}{7}$ **D)** **15**



13. The consecutive points **A, B, C, D** and **E** are on the same straight line where $DE=3AB, CD=2BC$ and $AC=8$. Find **BE.**

A) 8 **B) 16** **C) 24** **D) 32**

14. The consecutive points **A, B, C** and **D** are on the same straight line where $AB+CD=k$. If M and N are the midpoints of $[AC]$ and $[BD]$ respectively. Find **MN.**

A) k **B)** $\frac{k}{2}$ **C)** $\frac{k}{3}$ **D)** $\frac{2k}{3}$

15. The consecutive points **A, B, C** and **D** are on the same straight line where $AD=42$ and $AC+BC+2CD=70$. Find **AB**.

A) 14 **B) 12** **C) 16** **D) 13**

16. The consecutive points **A, B, C** and **D** are on the same straight line where $AB - CD = 22$. If M and N are the midpoints of $[AD]$ and $[BC]$ respectively. Find **MN.**

A) 16.5 **B) 5.5** **C) 18** **D) 11**

17. Simplify expressions $\frac{a^2-9}{5a^3b^3} \div \left[\frac{a+3}{10a^4} \cdot \frac{2a-6}{ab^4}\right]$.

A) ab **B)** a^2b **C)** $\frac{ab}{2}$ **D)** ab^2

18. If $1 \le k \le 25$. How many prime numbers are there in the form of $6k+1$?

A) 18 **B) 12** **C) 16** **D) 14**

19. Ratha reads the same book every day. The table below represents the total number of pages that the Ratha has read at the end of a given number of days.

Number of pages	
Number of days	Total number of pages
2	**32**
4	**64**
5	**80**
7	**112**

How many pages does Ratha read in one day?

A) 16 **B) 14** **C) 22** **D) 15**

20. A student has a bus ticket with a total value of $ 30.00. Every time a student takes a bus, the money on the bus ticket is deducted $ 2.25. How many times can a student ride the bus with the bus ticket?

A) 15 **B) 14** **C) 10** **D) 13**

21. One store sells blue hats and green hats. Each hat costs \$ 12.00. The expression $12m+12n$ can be used to determine the total price when a customer purchases the number of blue hats m and the number of green hats n. Which equivalent expression can be used to determine the total price in dollars of the hats?

A) $12mn$ **B)** $12(m+n)$ **C)** $6(m+n)$ **D)** $24mn$

22. Find the value of $\sqrt{1^3+2^3+3^3+\cdots+10^3}$.

A) 79 **B) 86** **C) 55** **D) 66**

23. Find the value of $(0.000064)^{\frac{2}{3}} \div (0.0016)^{\frac{3}{4}}$.

A) 5^{-1} **B)** 8^{-1} **C)** 4^{-1} **D)** 10^{-1}

24. Ratha is 51 years old and Bopha is 21 years old. How many more years will Ratha be twice as old as Bopha?

A) 10 **B) 12** **C) 6** **D) 9**

25. Sophea bought 5 kilograms of oranges, which were 10 oranges per kilogram. She ate eight fruits and gave the rest of her oranges to her six younger siblings. How many oranges did each of her siblings get?

A) 8 **B) 7** **C) 12** **D) 5**

26. Find the value of a and b such that $PGCD(a,b)=12$ and $a+b=72$ where $0<a<b$.

A) $a=32, b=12$ **B)** $a=24, b=60$ **C)** $a=12, b=60$ **D)** $a=12, b=24$

27. We know the pattern $\frac{1}{1\times2}=\frac{1}{1}-\frac{1}{2}$, $\frac{1}{2\times3}=\frac{1}{2}-\frac{1}{3}$, $\frac{1}{3\times4}=\frac{1}{3}-\frac{1}{4}$. As this pattern, the value of $\frac{1}{1\times2}+\frac{1}{2\times3}+\frac{1}{3\times4}+\frac{1}{4\times5}+\cdots+\frac{1}{2023\times2024}$ is:

A) 1 **B)** $\frac{2024}{2023}$ **C)** $\frac{2023}{2024}$ **D)** $\frac{1}{2}$

28. Sok and Sao invested **$12,100** in business. $\frac{3}{5}$ of Sok's share is equal to $\frac{7}{9}$ of Sao's share. How much is Sao's shareholding?

A) $5269 **B) $6,831** **C) $5,423** **D) $5,420**

29. Last week, the price of apple at a grocery store was **$1.60** per kilogram. This week, apples at the same grocery store are on sale for 10% off. What is the total price of 4kg of apples this week at the grocery store?

A) $6.93 **B) $6.75** **C) $4.77** **D) $6.48**

30. A worker arrives at work 15 minutes late as he walks at 4 kilometers per hour from home. The next day, he accelerates 2 km / h and arrives at work on time. Find the distance from his home to his workplace.

A) 5km **B) 3km** **C) 4km** **D) 2km**